Lemma γ is a constant speed geodesic if and only if

$$\frac{d}{dt}\left(g_{ij}\dot{\gamma}^j\right) = \frac{1}{2}\dot{\gamma}^j\dot{\gamma}^h\left(\partial_i g_{jh}\right) \quad \text{for } i=1,2$$

We can rewrite this using

$$\begin{aligned}\frac{d}{dt}\left(g_{ij}\dot{\gamma}^j\right) &= g_{ij}\ddot{\gamma}^j + \dot{g}_{ij}\dot{\gamma}^j \\ &= g_{ij}\ddot{\gamma}^j + \partial_l g_{ij}\,\dot{\gamma}^l\dot{\gamma}^j\end{aligned}$$

The equation becomes

$$\ddot{\gamma}^m = -g^{im}\left(\partial_l g_{ij}\right)\dot{\gamma}^j\dot{\gamma}^l + \frac{1}{2}g^{im}\left(\partial_i g_{hj}\right)\dot{\gamma}^j\dot{\gamma}^k$$

$\Rightarrow$ ODE system $\begin{cases}\dot{\gamma}^j = \eta^j \\ \dot{\eta}^j = f^j(\eta,\gamma)\end{cases}$

$\exists$ unique solution given $\gamma^j(0)$ and $\eta^j(0)$

Thm Given $p\in S$ and $X\in T_pS$, $\exists$ a unique geodesic with

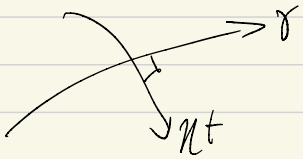
$$\gamma(0)=p \quad , \quad \dot{\gamma}(0)=X$$

Geodesic coordinate patch

Using coord's based on geodesic will make equations much simpler

Starting from $p\in S$, choose a reference geodesic γ with $\gamma(0)=p$, $\|\dot{\gamma}\|=1$

For each t, let η^t be a geodesic through $\gamma(t)$, $\perp$ to γ



$$\eta^t(0) = \gamma(t)$$

$$\dot{\eta}^t(0) = \dot{\gamma}(t)\times N$$

This defines a geodesic normal patch based on γ

$$\sigma(s,t) := \eta^t(s)$$

example: unit sphere, γ = equator

$$\sigma(s,t) = (\cos s \cos t, \cos s \sin t, -\sin s)$$

Properties of geodesic normal coord.'s

1) $g_{11} \equiv 1$ $\quad \sigma_1 = \dot{\eta}^t$ and η^t is unit speed

2) $g_{12} \equiv 0$

initial condition $\dot{\eta}^t(0) \perp \dot{\gamma}(t) \Longrightarrow g_{12}(0,t) = 0$

With t_0 fixed, $\eta(s) := \sigma(s, t_0)$ is geodesic

$$\Longrightarrow \frac{d}{ds}(g_{ij}\dot{\eta}^j) = \frac{1}{2}\dot{\eta}^i\dot{\eta}^k(\partial_i g_{jk}) \qquad i = 1,2$$

$i=2$:

$$\frac{d}{ds}(g_{21}) = \frac{1}{2}\partial_2 g_{11} = 0 \quad \text{since } g_{11} = 1$$

$$\Longrightarrow \frac{d}{ds} g_{21} = 0$$

$$\therefore g_{21} \equiv 0$$

4) $g_{22} = 1 + O(s^2)$ near $s=0$

γ unit speed $\Longrightarrow g_{22}(0,t) = 1$

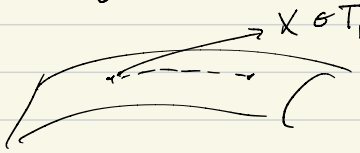
γ geodesic $\Longrightarrow \frac{d}{dt}(g_{12}(0,t)) = \frac{1}{2}\partial_1 g_{22}(0,t)$

Prop In geodesic normal coordinates $\sigma(s,t)$

$$g_{ij}(s,t) = \delta_{ij} + O(s^2) \quad \text{for all } t.$$

Geodesic polar coord's

The geodesic eq. allows us to map T_pS to S



define: For $X \in T_pS$, $\exp(X) := \gamma_{p,X}(1)$ where

$\gamma_{p,X}$ is geodesic with initial conditions p, X

If we fix orthogonal unit vectors E_1, E_2 in T_pS then we can define polar coord's

$$\sigma(r,\theta) = \exp\left(r\cos\theta\, E_1 + r\sin\theta\, E_2\right)$$

example: sphere, p = north pole

$$\sigma(r,\theta) = (\sin r\cos\theta, \sin r\sin\theta, \cos r)$$

$$ds^2 = dr^2 + \sin^2 r\, d\theta^2$$

Properties

1) $g_{11} \equiv 1$ (radial curves are unit speed)

2) $g_{12} \equiv 0$

radial curves are geodesic $\Rightarrow \frac{\partial}{\partial r} g_{12} = 0$

Computing at $r=0$ is delicate, so compare to Cartesian

Let $\tilde{\sigma}(x,y) = \exp(xE_1 + yE_2)$

$\tilde{g}_{ij}(0,0) = \delta_{ij}$ by choice of E_j

$$[g_{ij}] = \begin{pmatrix} \cos\theta & \sin\theta \\ -r\sin\theta & r\cos\theta \end{pmatrix} [\tilde{g}_{ij}] \begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix} \qquad (*)$$

$$\Rightarrow \lim_{r\to 0} [g_{ij}] = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$\Leftarrow$ $g_{12} \equiv 0$

3) $g_{22}(r,\theta) \sim r^2$ as $r \to 0$

follows from (2) also

Prop. In geodesic normal coordinates

$$ds^2 = dr^2 + \phi^2 d\theta^2 \quad \text{with } \phi \sim r \text{ as } r \to 0$$